

ENHANCING FINANCIAL INCLUSION : A TRANSPARENT AND FAIR CREDIT SCORING MODEL FOR RURAL POPULATIONS

Abstract

Purpose

Our research goal that utilizes Explainable AI (XAI) to develop a digital credit scoring solution that is meaningful for the people we serve is what our research team has done to improve upon what is already an excellent digital credit scoring system in rural areas of India.

Design/methodology/approach

We are going to utilize a combination of various techniques including standard scaling, one hot encoding and SHAP values from the several ensemble algorithms (XGBoost, Random Forest, and Logistic Regression). To make the application user-friendly for lenders, we will deploy a web-based interface using Streamlit. The models were evaluated on rural data to measure performance while keeping misclassifications to a minimum.

Findings

Utilization of XGBoost specifically decreased error rates, albeit providing lenders with more insight into which components influence their lending decisions through the illustration of SHAP values. For lenders, this is advantageous as it assists them in understanding the key factors influencing their various lending decisions.

Originality

The purpose of this document is to demonstrate solutions for addressing the issues within the area of rural lending, in which an overall lack of transparency and fairness exists. The examples provided illustrate how the application of ensemble learning techniques combined with the use of visual dashboards that are easily understood can impact rural lending practices.

Implications

We've developed a system that ensures fair credit evaluations. As a result, individuals who require credit can benefit from it while ensuring that the financing process encourages financial inclusion.

Keywords

Machine Learning; Financial Inclusion; Rural Finance; Credit Scoring; Explainable AI; Logistic Regression; Random Forest; XGBoost; Ensemble Models; SHAP; Streamlit

1. INTRODUCTION

Rural communities, which had traditionally been excluded from the formal credit market, are now benefiting from the digital lending industry which has expanded access to all financial services. However, even with many advancements, access to credit continues to be challenging for many rural borrowers. Most rural borrowers find it challenging to access credit because their

incomes are not stable and their credit histories are non-existent. Lending institutions continue to use the same standard lending approaches that were designed for urban areas; however, many rural borrowers are automatically rejected by lending institutions due to their inability to fit into the existing lending model, and from a tendency of lending staff to discriminate against borrowers based on location. Most lenders do not consider the unique aspects of rural creditworthiness, such as cyclical income from farming, different sources of business income that are available, and informal modes of doing business; thereby, excluding borrowers who otherwise would qualify.

We have been conducting research using machine learning methods to investigate this issue. So far, we have made significant advances; we've begun testing models that utilize various forms of applicant information (e.g., demographic data, digital payment history, landownership). The algorithms Random Forest and XGBoost appear capable of processing complex, multidimensional datasets related to rural areas; they can account for many different influences on each other and the non-linearities among them. The improvement these two algorithms provide over standard logistic regression or rule-based scoring method is substantial. We recognize, though, that although predictive accuracy is necessary, it isn't sufficient. In developing countries, or emerging markets, we need to demonstrate that those models are not only capable of making accurate predictions but also able to exhibit some degree of explainability.

The focus of this project is to create an explainable ML framework for evaluating creditworthiness to alleviate issues associated with unfairness and unethical lending practices in Rural India. This framework develops and utilizes clean and meaningful inputs for creditworthiness leveraging one-hot encoding and standardizing the data. Feature engineering (e.g., selecting) will identify which factors are most relevant or predictive for lending decisions. Additionally, the use of SHAP graphs/evaluations facilitate understanding of how the model made its decision (Lal and Kumar, 2024). The Risk Score Dashboard (developed with Python & Streamlit) provides banks and lenders an opportunity to review Risk Scores in real-time and view geographic allocations of risk scores, allowing them to accurately categorize rural borrowers and make funding decisions appropriately (Ghimirey et al, 2025). Model testing conducted on actual rural datasets demonstrated a reduction in incorrectly classified loans, which increases equity for borrowers who may have been overlooked with previous scoring models (Kumar and Singh, 2022). However, the explainable ML framework has inherent limitations. It requires the continual enhancement of the model to accommodate rural populations across multiple geographies, making it essential that digital records be maintained as reliable, accurate sources of customer information (Ghimirey et al., 2025). Nevertheless, this model has demonstrated tangible improvements in Rural Creditworthiness Assessments, demonstrating that by incorporating the proper preprocessing techniques, implementing an explainable ML framework, and employing visual aids to better understand the creditworthiness assessment process, the model produces a greater level of fairness within the Financial Services Industry (Lal and Kumar, 2024).

2. REVIEW OF LITERATURE

In this study, (Kumar *et al.*, 2021) Discusses how machine learning (ML) is utilized to Digital Credit Scoring (DCS) in the context of rural financing. To address the challenges that exist in rural financial markets, the research discusses various technological advancements made within this field and identifies outstanding challenges yet to be addressed. As a result, the research provides clear evidence that traditional methods of assessing an individual's creditworthiness (i.e., 5Cs or other methods based on rules) do not include those individuals who lack formal documentation supporting their income. Due to this, small farmers and rural entrepreneurs have difficulty obtaining loans (Yu *et al.*, 2015). The authors further indicate that machine learning algorithms, including Decision Trees (DT) and Ensemble Learning Algorithms (ELA), enable lenders to better estimate the likelihood of repayment by using both traditional financial information and other alternative data sources (i.e., mobile payment data and farm record). (Ampountolas *et al.*, 2021).

From (Kumar *et al.*, 2021), several major challenges currently prevent the optimal utilisation of ML within rural environments. One of the main obstacles has been the low quality of rural datasets, which typically contain unbalanced class distributions, missing values and under-representation of digital records, resulting in diminished effectiveness for any ML model, no matter how sophisticated (Afonso Fontes and Gay, 2021). The authors of the reviewed studies concluded that 'standard models' such as logistic regression performed significantly better when large quantities of well-structured data were available, whereas advanced models, like neural networks and ensemble approaches, were more successful at assigning probability estimates to predictions, yet offered less insight as to the reasoning behind the conclusions drawn (Boughaci and Alkhawaldeh, 2018; Antunes, 2021). A number of the empirical investigations highlighted in the review (e.g., Aniceto *et al.*, 2020; Assef and Steiner, 2020) believe that hybrid/ensemble ML approaches outperform simple classifiers within the context of both microfinance and rural datasets when a solid feature engineering approach is taken. The review further states that recent fintech startups have been rapidly increasing their role within fin-tech, developing new ways to collect, analyse and offer lending to rural customers by leveraging digital channels and non-traditional/alternative sources of data, while still facing challenges such as the incorrect implementation of saved algorithmic performance metrics/decisions (Criado and Such, 2019) as well as potential threats to data privacy.

The combination of robust feature selection, data preprocessing, and SHAPs interpretability in the interactive dashboard directly addresses the gaps in the literature identified by previous studies. Unlike previous methods, this approach treats both accuracy and transparency equally, thus providing lenders with the tools they need to make safer lending decisions while also alleviating many of the ethical issues that were highlighted but not fully resolved in previous research (Abuhusain, 2020; Goh *et al.*, 2020). The dashboard is user-friendly, reusable, and

expandable to accommodate different rural communities, thus increasing the practical utility of credit scoring by lending organisations.

3. METHODOLOGY

The steps outlined in the Methodology section below serve as a structured, multi-phase process for creating and validating a strong credit scoring model. Figure A summarizes the entire workflow visually. The overall process described in the Methodology section is similar to that outlined in the Introduction. However, the technical methodology starts with the basic steps associated with acquiring data.

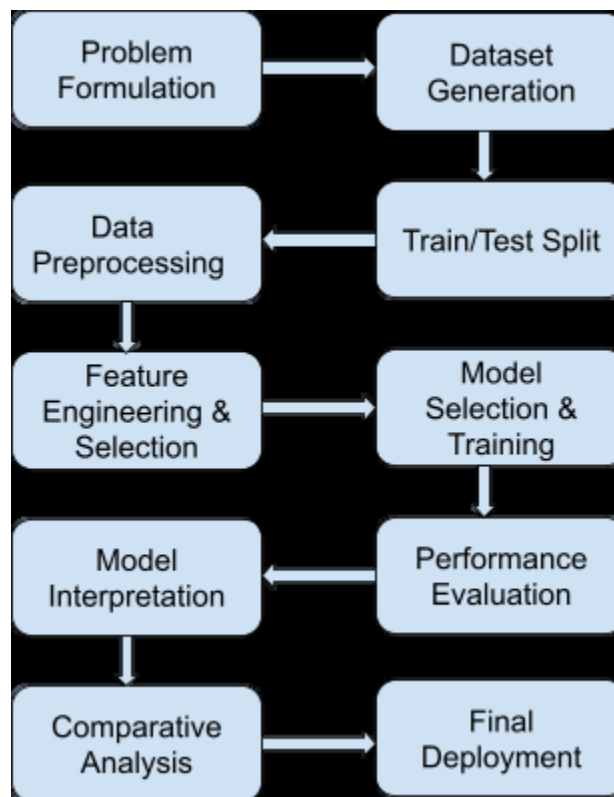


Figure 1 Flowchart

3.1 Dataset Description

In this study, datasets used for analysis were created using Python. These scripts were based on current rural India's agricultural and economic settings, and as a result, they generated realistic borrower data. The datasets contain 5,000 records with 23 attributes that provide the agricultural characteristics, economic factors, repayment behaviours, and demographic data of borrowers. The generation method provided a fair representation of the different types of borrowers in rural

India; consequently, this allowed for the creation of accurate models for predicting credit risk and effective engineering of data features (Kumar *et al.*, 2021). Prior research has shown that synthetic datasets that reflect characteristics of a particular industry or field can be utilised within finance to study risk; this is especially useful when all data is unavailable or difficult to obtain (World Bank, 2022; Kamar and Mishra, 2022). Many researchers studying rural finance employ similar types of simulations to allow for maintaining analytical validity while also providing protection of individual borrowers' privacy (Kaur *et al.*, 2023; Mishra and Swain, 2021).

3.2 Data Preprocessing

We preprocessed our rural credit simulation dataset and approved it for trustworthy machine learning modelling. We also removed any unnecessary client ID numbers (Agbo *et al.*, 2021), and retained only important information about borrowers and loan information. We used zero imputation to fill in the non-existent numerical values (such as yield or income). This method works well with structured financial datasets, particularly when they contain limited amounts of information (Kumar *et al.*, 2021). We converted categorical variables (such as gender and education) to string format, following the guidelines of Mestiri (2024), and created one-hot encoded string variables to provide uniformity in the format of our data in a machine-readable way. We divided the dataset into two sets: an 80/20 split for train/test datasets, ensuring the distribution of the target variable class remains unchanged and the validity of the evaluations performed on this data, and we used StandardScaler to standardise (or normalise) all continuous data (Kumar & Singh, 2022; Singh & Shivaprasad, 2023).

3.3 Feature Engineering and Selection

To improve learning and provide a clear indication of the project's success; feature engineering included both modification and creation of variables. In addition to using new variables created from the comprehensive borrower data set, such as Total Family Dependency Ratio (Dependents/Family Size) and Loan-to-Income Ratio and binary codes for past defaults/Divergents, existing variables like age, land size and income were retained. Non-numeric patterns can be recognised through categorical attributes, e.g. Crop Type, Irrigation Type, and State, which were subject to one-hot encoding techniques (converting categorical variables).

The use of feature selection through the application of tree-based models, particularly XGBoost, has been driven by domain knowledge and measurable value as it relates to creating a model through identification of the features that contribute the most to prediction. In order to ensure that models would be generalizable to different situations (to mitigate against over-fitting), regularization methods (which promote "shrinkage" of weights) combined with permutation analysis were used to exclude features that were highly correlated with other features and those that were minimally informative.

SHAP plots clearly illustrate features such as annual income, number of prior delinquencies, size of the property, and geographic location as the most significant predictors of the likelihood of repaying loans. All the mentioned processes resulted in a trained and validated machine-learning (ML) pipeline that has good interpretability versus high performance.

The inclusion of the SHAP feature importance plots supports these decisions, with the SHAP summary plot representing a visual representation of the ranking in importance of all features impacting credit-scoring predictions in Figure 2.

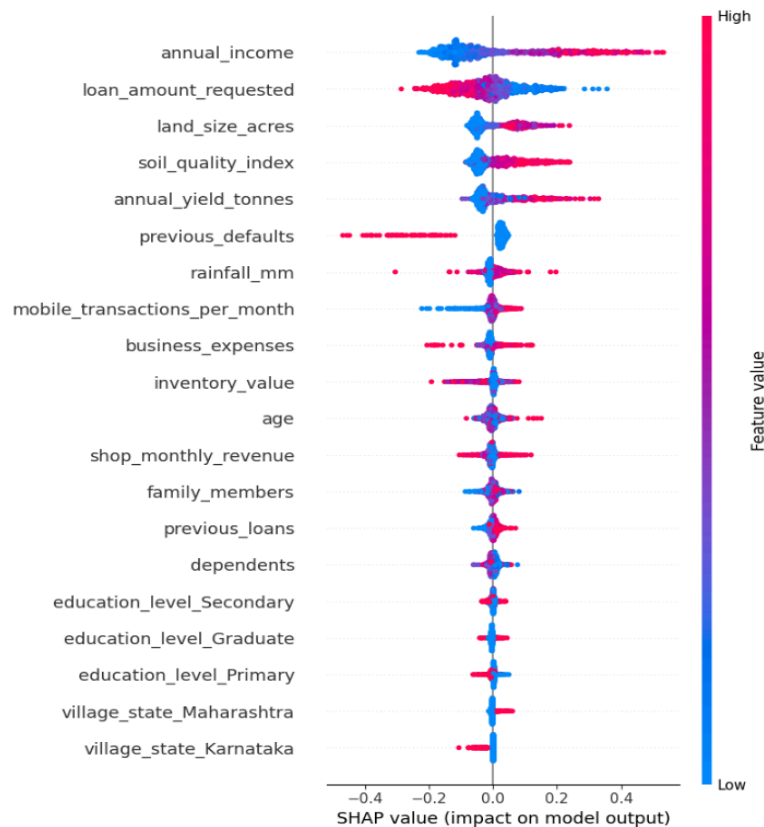


Figure B SHAP value

3.4 Model Selection and Training

For this project, when it came to model selection and training, two major priorities for us were interpretability and predictability. The first priority is important when evaluating rural lending credit risk. We conducted an assessment of multiple supervised classification models, including the very powerful XGBoost framework; Random Forest; and Logistic Regression.

I. Logistic regression

Logistic regression operates under the assumption that the logarithm of odds for loan repayment can be modeled as a linear combination of input features. The decision function is (Bishop, 2006):

$$P(y = 1 | X) = \frac{1}{1 + \exp(-(\beta_0 + \beta_1 x_1 + \dots + \beta_n x_n))} \quad (1)$$

Where $P(y = 1 | X)$ is the probability of loan repayment, β are the coefficients learned from the data, and $x_1, \dots, x_n$ are input features.

II. Random Forest

An ensemble of decision trees called Random Forest is created where every tree T_i gets trained with a bootstrapped sample. The ultimate result of the predictions will be determined by majority vote or average for classification and regression purposes, respectively (Breiman, 2001):

$$\hat{y} = \frac{1}{K} \sum_{i=1}^K T_i(X) \quad (2)$$

Where K is the total number of trees.

III. XGBoost

The way that XGBoost works is that it uses a gradient boosting algorithm to train an ensemble of "weak" classifiers, usually decision trees, in order to create a better predictive model by adding a new weak classifier (i.e., decision tree) on each iteration of the boosting process. Whenever a weak classifier is added, it is trained to predict the negative gradient (the difference between the actual and predicted value at the previous iteration) of the loss function with respect to the current set of predicted outcomes. The XGBoost model is made up of several "small" trees (the f_k trees) which together form the overall predictive function f_k (Chen and Guestrin, 2016):

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i), f_k \in F \quad (3)$$

Here, F is the space of regression trees. The objective function optimized in XGBoost for n training examples is (Chen and Guestrin, 2016):

$$L = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (4)$$

Where l is the loss function (e.g., logistic loss for classification), and Ω is a regularization term that penalizes tree complexity (Chen and Guestrin, 2016):

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda ||w||^2 \quad (5)$$

The number of the leaves T , hyperparameters γ and λ on their own, and the weights of the leaves w were fixed before initiating training. The split of the input dataset into training and testing subsets was 80/20. Training of the XGBoost model was achieved by searching the best set of hyperparameters (number of trees, tree depth, learning rate) that would avoid overfitting and improve generalization.

The evaluation of our models incorporated Accuracy, Precision, Recall and F1 scores; we also plotted ROC curves to illustrate true and false positive rates (Dastile *et al.*, 2020). We defined Feature Importance through XGBoost Gain and SHAP Value Metrics to allow for easy interpretation of our final model in the Credit Scoring Applications (Jha & Pandey, 2022). This procedure resulted in a strong classifier with nuanced risk assessment capabilities, allowing for appropriate performance and transparency of reasoning for Digital Rural Lending.

3.5 Model Validation and Evaluation

Use a variety of standard model validation and evaluation metrics to provide an accurate reflection of smallholder loans classification algorithms and their performance against a rural credit scoring application by addressing the following evaluation metrics : accuracy, precision, recall (sensitivity), F1 score , and Area under the ROC Curve (AUC) , which measure how well the model predicts smallholder loans to farmers from different perspectives. The accuracy metric indicates how many predictions by the model were classified as correct.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (6)$$

(Sokolova and Lapalme, 2009)

Precision indicates how many positives are actually positive from all the predicted positives:

$$Precision = \frac{TP}{TP + FP} \quad (7)$$

(Sokolova and Lapalme, 2009)

Recall checks the model's sensitivity in capturing actual positives:

$$Recall = \frac{TP}{TP + FN} \quad (8)$$

(Sokolova and Lapalme, 2009)

F1 score is a performance measure for a model that gives one single number that combines precision and recall via their harmonic mean:

$$F1\ Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (9)$$

(Sokolova and Lapalme, 2009)

The ROC curve represents the variability of True Positive Rate (TPR) with the variations in False Positive Rate (FPR) at varying threshold values (Fawcett, 2006).

$$TPR = \frac{TP}{TP+FN} \quad (10)$$

$$FPR = \frac{FP}{FP+TN} \quad (11)$$

The metrics above provide an opportunity for more thorough comparison of the models compared to one another, thus enabling the identification of the algorithms that are most suitable for making reliable risk assessments in digitally-enabled rural lending environments.

3.6 Model Deployment

A streamlit web application was developed as the last machine learning pipeline i.e. preprocessing, feature engineering, model interface, SHAP explanation. With this interactive Streamlit web application, credit scores can be produced in real-time according to a rural lending situation. Streamlit provides several advantages through its lightweight structure and ability to create applications rapidly, as well as interoperability with any Python-based Machine Learning ecosystem. Streamlit app provides lenders with easy-to-use features allowing lenders to enter data on borrowers, including demographics, agricultural data, financial indicators and past loans. After entering the data into the application, the application passes this information through a trained XGBoost Model to generate the credit score and provides corresponding explainable SHAP visualizations. Thus, lenders can see how the prediction was generated and identify the important variables. The Digital Credit Scoring system that was deployed (shown in Figure 3) offers many features that help lenders be accurate, easy to use and transparent, when assessing rural borrowers in real-time.

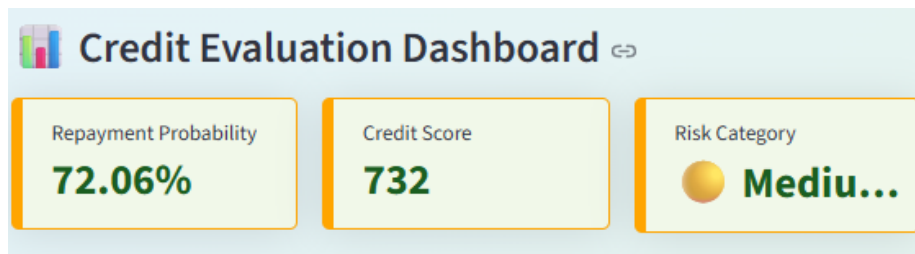


Figure 3 Dashboard

The Real-time Score is within the Pipeline of the trained machine learning model. When an applicant submits new applicant information it will be scored for credit risk immediately. The real-time scoring process automatically preprocesses applicant's information. The real-time scoring is done using the transformed data with scaling, categorical encoding and feature transformation from the original data. Figure 4 shows the top Predictive Factors that contribute to determining an individual borrower's ultimate Credit Score through SHAP.

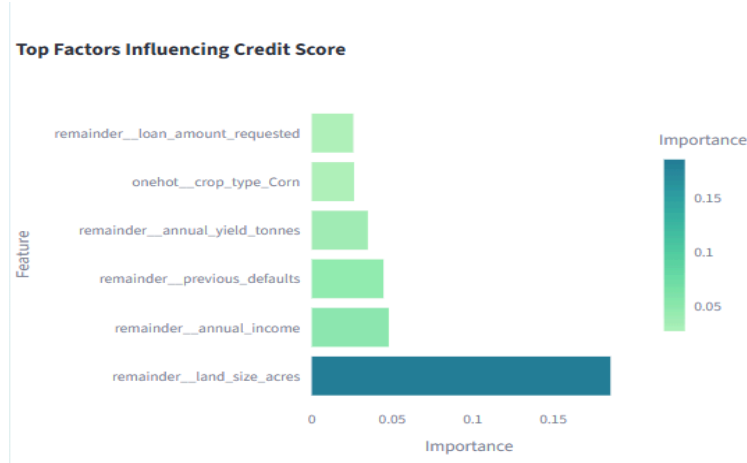


Figure 4 Top factors influencing credit score

4. EXPERIMENT AND RESULTS

4.1 Experimental Setup

We evaluated three of our top-performing classifiers, (Logistic Regression, Random Forest, and XGBoost), using a custom-built dataset for rural borrowers specifically created for our research. After performing multiple preprocessing tasks (such as replacing missing data and converting categorical variables into numeric), we divided the dataset into two sections, an 80% training section and a 20% testing section. To confirm that our experiments would yield reproducible results and that the tests would not reflect any random bias from a single run of the test, we established fixed random seeds for all three classifiers and also utilized XGBoost's grid search tuning capabilities. Our classifiers were then subjected to several popular classification tests to evaluate their actual performance on this dataset.

4.2 Performance Metrics

We assessed the overall effectiveness of each model by calculating its accuracy - essentially, the number of times it was able to successfully classify instances. We also evaluated each model's precision and recall to gain a sense of how well they performed with respect to classifying the probability of an instance being classified as a borrower who will repay or won't repay as well. If there was an imbalance in the dataset with respect to the number of instances of different types of borrowers, we calculated the F1 score to account for this imbalance by combining precision and recall as a single value. In order to provide more insights into how well our models performed at varying thresholds, we generated ROC curves and calculated AUC values for each of the models. Finally, we produced a confusion matrix for each of the models detailing the number of true positives, true negatives, false positives, and false negatives. Combining this information provided an overall picture of how effective our digital lending solution was and helped us identify which credit scoring model performed best and how they compared.

4.3 Results and Comparative Analysis

The ROC Curve for Logistic Regression from our proprietary Rural Credit Data set presents us with a solid indication regarding the accuracy of this fitting; it has decent separation capability due to the AUC value being apparent, thus illustrating both the strengths as well as limitations of the Logistic Regression Model. In addition, Figure 5 displays the ROC Curve that corresponds with the Logistic Regression analysis; hence we can gain insight into the relationship between true positive and false positive errors.

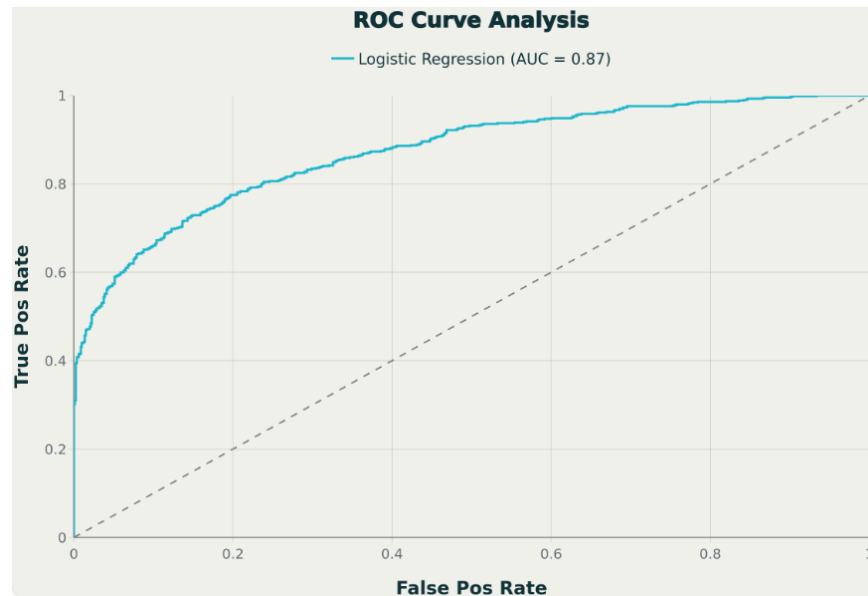


Figure 5 ROC curve

From the results it's evident that although logistic regression classifies borrowers who repay their loans as opposed to those who don't reasonably well; XGboost has shown to outperform all other classification models at generating accurate predictions. In this example XGboost has produced superior ROC-AUC results compared to all other classification models while logistic regression follows closely behind, while Random Forest has produced lower levels of consistency with respect to both measures. Comparing these three classifiers indicates that when assessing credit risk associated with rural lending, that ensemble modelling technique has demonstrated its highest level of recommendation and should be considered prior to making any credit decisions.

4.4 Interpretation of Results

The information provided in Table I. lists the KPI's for three models that will play an important role in predicting rural lending defaults. First up, Logistic Regression is great because it is simple to understand and is intuitive to use. However, it is not very accurate (0.68 accuracy, ROC-AUC 0.68) when it comes to differentiating between borrowers who pay back their loans on time from

those who do not. Next, the Random Forest model is a significant improvement over Logistic Regression because of its increased accuracy (0.77 accuracy, ROC-AUC 0.76), meaning that Random Forest can model more complex relationships among the different features present in the data. The downside to Random Forest is that it is still having trouble detecting all of the borrowers, including those who have defaulted.

Through our analysis of many different machine learning algorithms to do with rural credit risk analysis, XGBoost clearly stands out above all other options. The numbers show that XGBoost is the top performer in nearly every performance metric; it has the highest overall accuracy and precision scores at 0.86 and 0.93, respectively. Additionally, XGBoost has one of the strongest ROC-AUC scores (0.89) when compared with other machine learning algorithms. This means that when trying to predict which individuals will default on their rural loans and which ones will pay their loans in full, XGBoost can predict with accuracy the probabilities of each borrower paying or defaulting on their loan; therefore, XGBoost has a clear advantage for performing rural credit scoring correctly. Based upon these results, it appears that utilizing ensemble learning techniques, such as XGBoost, should provide the best possibility to obtain high accuracy and precision with your rural credit scoring system. As such, we would definitely recommend XGBoost, based upon our analysis, as the best algorithm for performing credit scoring analysis within the context of rural loans to find the most suitable/qualified borrowers. Random Forest, although rated fairly well in the middle of our evaluation, could not be considered nearly as effective as XGBoost at this time; whereas Logistic Regression would not be an effective means of analyzing highly complex borrower repayment behaviour. Overall, when assessing which machine learning algorithm to use to evaluate credit risk for rural loans, we feel XGBoost is the most appropriate algorithm to consider using.

[Insert Table I here]

Table I : Comparative Model - performance Metrics

5. DISCUSSION

Machine learning-based digital credit scoring systems, which offer the capability of assessing risk and being more transparent about lending decisions than traditional credit-based scoring methods, can significantly boost the availability of capital to rural communities. Previous research has shown how traditional lending practices for most lenders leave rural borrowers without access to credit because these borrowers usually lack any established credit history or reliable income, which lenders typically rely on to evaluate a borrower's ability to repay (M&J, 2024; Ding *et al.*, 2025). Hand in hand with new statistical models such as logistic regression, random forest and particularly XGBoost, these alternative lending methods can take advantage of other sources of available data unique to rural areas. Based on this type of data, accurate and easily-understood credit scoring algorithms can be developed (Dastile *et al.*, 2020; Zekri *et al.*,

2021). The use of aggregation in ensemble methods of learning leads to consistently superior models for rural lending than any other form of modeling through including agriculture, sociology and demographics as part of their feature engineering (Trivedi and Kumar, 2020; Naqvi *et al.*, 2022). Further, the regularized objective function and gain-based feature importances of XGBoost provide at present the most consistently predictable way of evaluating rural borrowers (Kumar *et al.*, 2021; Melese *et al.*, 2023).

This project kept our models simple by using logistic loss structure, while also prioritizing model selection and training, to maximize predictive power and provide interpretability of model outcomes. This correlates with the conclusions being found in the literature regarding the best practices for developing rural credit models (Oyeyemi *et al.*, 2025; Sharma & Priya, 2025). Streamlit's reactive interface is another tool that has given lenders, rural clients, and any other potential user of this new technology easier access to credit scoring tools. The application gives the user the ability to quickly access a credit scoring tool without needing to be an expert user of any particular software (M&J 2024; Wang, Li, *et al.*, 2022). It uses pre-trained machine learning algorithms on the back end of its application to provide the user with a very user-friendly way of interacting with these algorithms, with features such as forms, sliders, and on-demand visualizations to help users better understand the algorithms they are using. This application will directly support efforts to resolve the issues of lack of transparency and lack of trust in lenders and loan applicants (Jha & Pandey 2022; Shreya & Pathak 2025). The use of straightforwardness and clarity are critical when it comes to success in increasing the number of people who have access to financial services and therefore improving financial inclusion. As noted by (Kumar *et al.*, 2021; World Bank 2022; Kamar and Mishra 2022), this continues to be true according to our findings as well as for the fintech sector. Additionally, recent research has shown that the implementation of Streamlit's digital scorecarding technology by (Singh and Shivaprasad 2023) and (Li *et al.*, 2022) enables the lending decision-making process to be performed in a manner that is both equitable and bias-free, and on a large-scale basis, for rural loan applicants. Furthermore, not only does this platform enhance and streamline regulatory compliance, it also allows the user to continually update and modify their data collection methodology; and it maintains an ethical lens; in essence, the implementation and expansion of each of these methodologies can take place in real-time as well as the development of policies that are currently under discussion in terms of appropriate and responsible ways to evaluate credit applicants (Wang *et al.*, 2023 & Vasudevan 2022). As a result, the combination of leading-edge artificial intelligence with state-of-the-art web-based solutions, which allow the users to be engaged with the products, is the key to developing a comprehensive solution i.e., an ethical and user-centric rural credit evaluation process.

6. FUTURE SCOPE

The next step for this research is to use authentic borrower data from rural communities to determine whether or not these models are accurate in real-world lending environments where

there are more complex real-life considerations. Furthermore, by incorporating newer variables that are even more relevant in regards to how a farmer operates on a daily basis such as changes in weather patterns, fluctuating prices for local commodities, or simply providing geospatial reference points for transportation costs and isolation, many of these “small” variables can greatly change the way in which a farmer will calculate his/her risk potential. The next challenge is to develop models that are more transparent to customers regarding the decision-making process. There are several different tools available to the public, including SHAP, LIME, and even counterfactual explanations, which may allow customers and lenders to view how the model is interpreting their information and provide information to support its claim, but none of these tools can tell customers how to achieve cause and effect. Therefore, it is important for both lenders and borrowers to be able to utilize the tools provided. This model has the potential to develop into a larger scale digital CSF model with numerous use cases. For example, the model may provide a digital financial platform that incorporates automated assessments of loan applications and an adjustable loan pricing schedule based on an assessment of risk. Although automation is beneficial, it will only be possible if there is an ongoing trust in the model being used for both lenders and borrowers. Therefore, to allow for a trust flow, the model will need to be regularly monitored and re-trained or possibly redesigned to adapt to changes in the regulatory environment/economic landscape/cultural changes.

7. CONCLUSION

This study uses four machine-learning techniques (Random Forest, Logistic Regression, and XGBoost) to build a model for predicting which rural borrowers will repay a loan. The results indicate that the combination of models outperforms the individual techniques as a predictor of safe or risky borrowers, especially the XGBoost model, which effectively combines socio-economic and demographic data with agricultural data. The XGBoost model handles the challenges of the combination of socio-economic, demographic and agricultural variables quite well and is resilient to the difficulties presented by new and limited rural data. This is very advantageous for rural lending, as many data in rural areas are often messy and do not have clear patterns. In addition, we provided the lender with an easy-to-use tool (SHAP) that provides the lender with the explanation behind model predictions, allowing lenders to better understand, question and trust the model's decisions. Although it does not address every concern about the lender's interpretations of the model, SHAP allows for a more straightforward understanding of the decision-making process. The system also meets critical ethical and regulatory expectations for financial risk assessment. In addition, we used a simple Streamlit application interface, making it easier for lenders to interact with the model and use it without technical expertise. Users have the ability to navigate through the selections, view intuitive and clear graphs, and now have the option to check their credit score at any time. The analyses performed during testing with respect to the three different models clearly demonstrate that the performance of each model varies. Initial results indicated that Logistic Regression performed poorly due to its inability to provide straight-line classifications only and therefore had a ROC-AUC score of

0.68, which, while acceptable, is not suitable given the random and unbalanced nature of the datasets typical for rural regions. Random Forest performed marginally better than Logistic Regression with respect to ROC-AUC score of 0.76 and demonstrated the ability to account for more complex behaviours within the dataset. However, XGBoost produced the highest ROC-AUC score of 0.89. XGBoost also surpassed all other models with the greatest overall accuracy, precision, and recall. The SHAP results provided insight into the most important variables contributing to credit risk assessment; Land size, Income Level and Past Loan Defaults. Overall, the Streamlit interface validated the fact that this platform provides lender users a smart and useful way to quickly access accurate credit risk assessments in real-time, resulting in lenders making more equitable lending decisions. Thus, the results indicate that Machine Learning has an opportunity to transform the way lenders determine the level of risk associated with extending credit. By minimizing these errors, these tools provide a more level playing field for those that wouldn't normally qualify for credit. Machine learning-based technologies have the potential to revolutionize the way lending decisions are made in rural communities that have long been neglected from traditional sources of loan funding. Although machine learning does not solve every systemic issue affecting rural lending, it does present an opportunity for greater digital inclusion into more practical, usable solutions for those that have been historically deprived of the opportunity to obtain reliable credit.

8.LIMITATIONS

The data set created to represent individuals borrowing in rural areas has flaws. Even though designed to resemble actual rural borrowers, the data set is still synthetic and does not represent the true representation of the credit behavior and rural lending practices of individuals living in rural areas. Secondly, this model does not include some of the critical behavioral, socio-economic and Geospatial risk factors that affect rural lending and therefore may miss capturing key risk factors. In addition, it is expected that models will differ based on the regional differences associated with agricultural practices and regional economies. Further, interpreting SHAP explainability methods is challenging, as these methods provide a general understanding rather than true causal insights. Finally, implementing this type of technology in a rural area's credit lending environment will be further complicated by limited access to digital infrastructure and compliance with regulatory requirements.

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